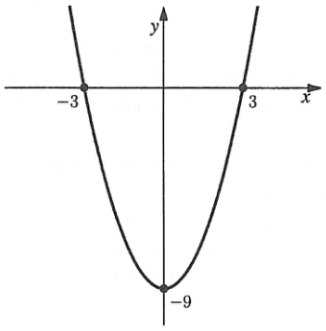
Graph Sketching: Quadratic Graphs

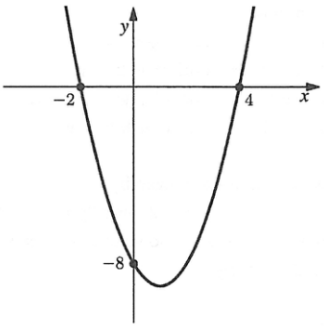
Questions 1–20

Question 1. Find the equation of each of the following graphs in the form $y = (x + a)(x + b)$

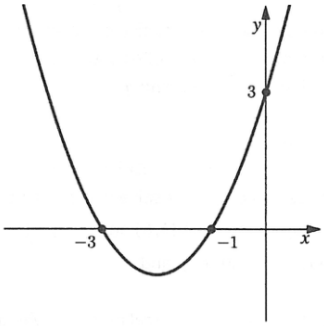
a)



b)



c)



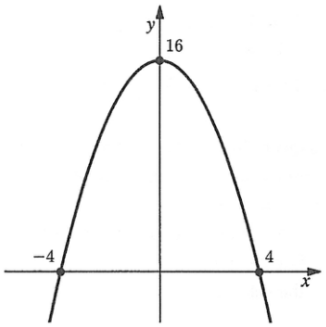
Question 2. Sketch the following graphs, showing where they intersect the coordinate axes:

a) $y = (x - 5)(x - 7)$

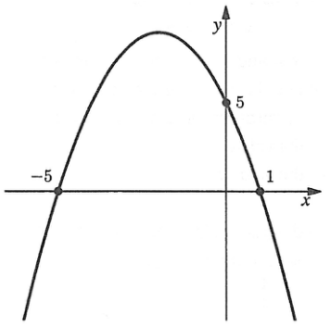
b) $y = (x + 4)(x - 2)$

Question 3. Find the equation of each of the following graphs in the form $y = -(x + a)(x + b)$

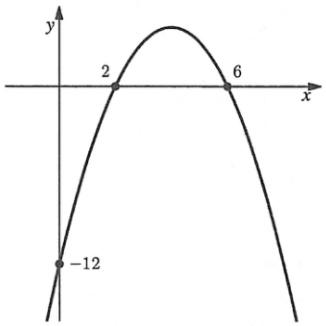
a)



b)



c)



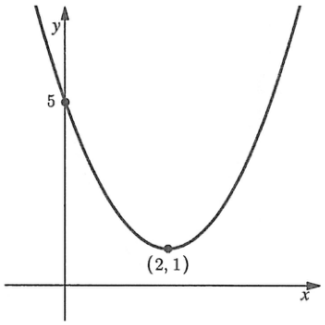
Question 4. Sketch the following graphs, showing where they intersect the coordinate axes:

a) $y = -(x + 2)(x - 5)$

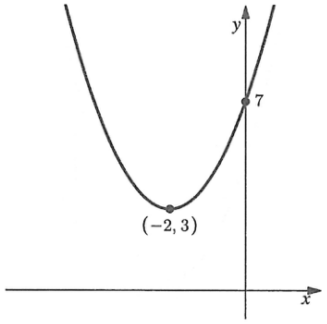
b) $y = -(x + 8)(x + 1)$

Question 5. Find the equation of each of the following graphs in the form $y = (x + a)^2 + b$

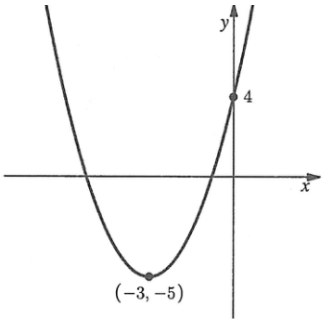
a)



b)



c)



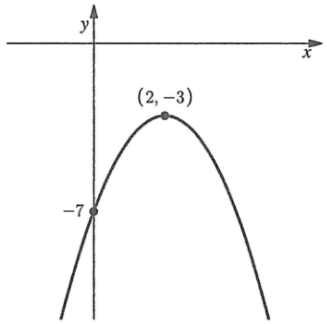
Question 6. Sketch the following graphs, showing the coordinates of the turning point and where they intersect the y-axis:

a) $y = (x - 4)^2 + 5$

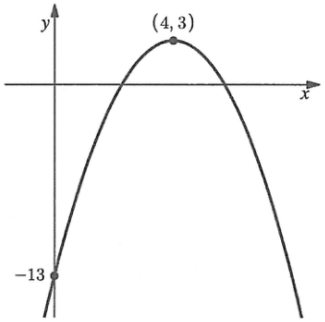
b) $y = (x - 5)^2 - 6$

Question 7. Find the equation of each of the following graphs in the form $y = -(x + a)^2 + b$

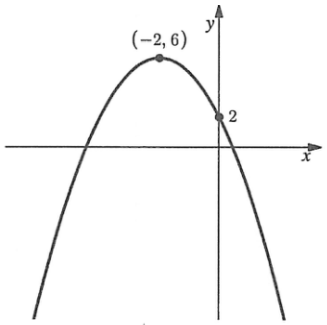
a)



b)



c)



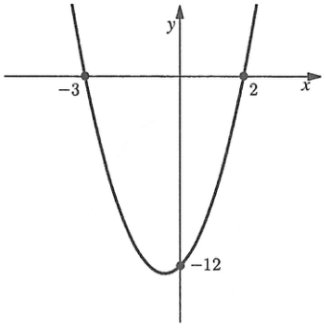
Question 8. Sketch the following graphs, showing the coordinates of the turning point and where they intersect the y-axis:

a) $y = -(x + 6)^2 + 2$

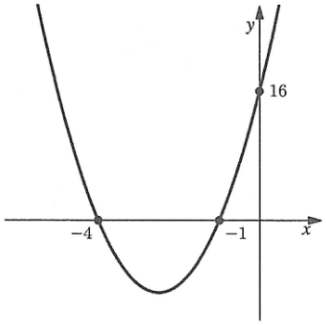
b) $y = -(x - 3)^2 - 4$

Question 9. Find the equation of each of the following graphs in the form $y = a(x + b)(x + c)$

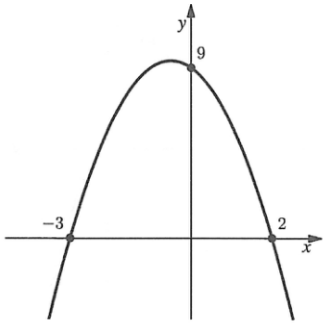
a)



b)



c)



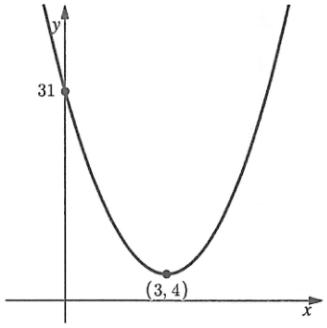
Question 10. Sketch the following graphs, showing where they intersect the coordinate axes:

a) $y = 3(x - 1)(x + 2)$

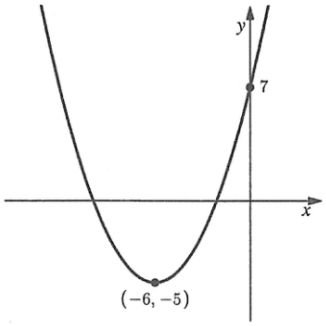
b) $y = -2(x + 4)(x + 3)$

Question 11. Find the equation of each of the following graphs in the form $y = a(x + b)^2 + c$

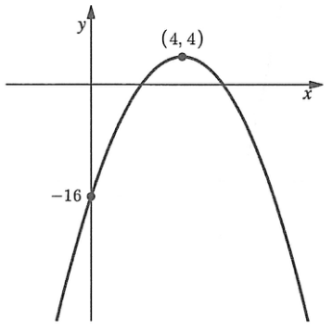
a)



b)



c)



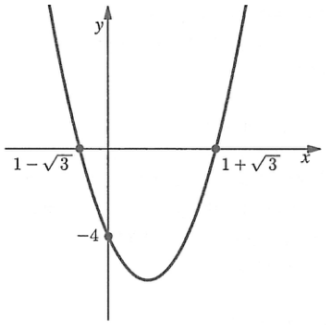
Question 12. Sketch the following graphs, showing the coordinates of the turning point and where they intersect the y-axis:

a) $y = 5(x + 1)^2 - 2$

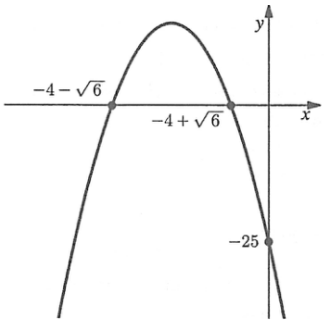
b) $y = -\frac{3}{2}(x - 6)^2 - 12$

Question 13. Find the equation of each of the following graphs in the form $y = ax^2 + bx + c$

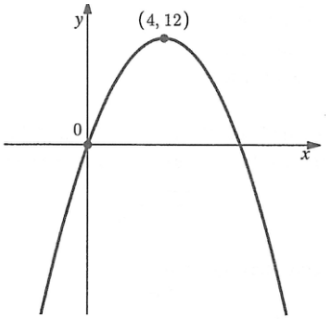
a)



b)



c)



Question 14. Sketch the following graph, showing the coordinates of the turning point and where it intersects the y-axis: $y = \frac{1}{2}(x+8)^2 - 32$.

Question 15. Sketch the following graphs, showing the coordinates of the turning point and where they intersect the coordinate axes:

a) $y = (x + 2)(x - 8)$
b) $y = (x - 6)(x + 4)$
c) $y = 3(x + 6)(x + 2)$
d) $y = -4(x - 2)(x - 4)$

Question 16. Sketch the following graphs, showing the coordinates of the turning point and where they intersect the coordinate axes:

a) $y = (x - 3)(x - 7)$
b) $y = \frac{1}{3}(x + 2)(x + 6)$
c) $y = -(x + 9)(x + 5)$
d) $y = -\frac{7}{4}(x - 8)(x + 4)$

Question 17. Sketch the following graphs, showing the coordinates of the turning point and where they intersect the coordinate axes:

a) $y = (x-6)^2 - 2$
b) $y = 8 - (x+2)^2$
c) $y = \frac{4}{3}(x+6)^2 - 9$
d) $y = -\frac{8}{3}(x-3)^2 + 12$

Question 18. Sketch the following graphs, showing the coordinates of the turning point and where they intersect the coordinate axes:

a) $y = 3x^2 - 9x + 2$
b) $y = -2x^2 + 10x + 2$
c) $y = \frac{1}{2}x^2 - 6x + 9$
d) $y = -\frac{1}{5}x^2 + 2x + 10$

Question 19. Given that $k > 0$, sketch the following graphs, identifying the turning point and where they cross the coordinate axes in terms of k:

a) $y = k(2x + k)(x - 2k)$
b) $y = k - \frac{1}{3}(x - 3k)^2$
c) $y = 10 + \frac{2}{k}(kx + 1)^2$
d) $y = (3kx + 2)(3k - x)$

Question 20. Given that $k > 0$, sketch the graphs of $y = (k - x)(3x + k)$ and $y = k - x$ on the same axes, showing clearly where the two graphs cross the coordinate axes and where they intersect in terms of k.